

Commonsense Psychology and Language

By the time we have become fluent speakers of our native languages, we have learned to use thousands of words and phrases to refer to mental states and processes of ourselves and others. This richness in vocabulary parallels the complexity of the commonsense psychological model that defines the deep lexical semantics of these linguistic expressions. By studying the richness of psychological language, we learn about the various facets of the knowledge representation challenge, and its overall breadth of scope.

In this chapter, we attempt to catalog the full breadth of English words and phrases related to commonsense psychology. We organize this vocabulary into twenty-nine representational areas, and group words and phrases into lists of entries that we consider to be conceptually synonymous. Undoubtedly we have missed quite a few expressions, and we have not attempted to list every synonym or morphological variation. Instead, our goal is specifically about breadth of coverage and granularity of conceptual distinctions. In creating these word lists, we are defining the target of our knowledge representation efforts. Our task is to develop a set of formal theories of commonsense psychology in which all of these words have a referent, and in which the theories themselves are rich enough to distinguish between each enumerated concept. We concede that there are subtle differences between words and phrases that we have grouped into a single enumerated concept. However, we believe that the concept categories capture much of the meaning of each group of expressions, and give us a good starting point for future efforts to articulate (formally) subtle differences in meaning.

In offering this (long) catalog here, we hope to achieve several goals. First and foremost, we can think of no better way to show readers the enormous breadth of commonsense psychology in terms that are already well understood. Second, we see these lists as a non-technical index to the topics that are formalized in Part III of this book. Where readers are interested only in certain topics of commonsense psychology, one need only to find the relevant concepts in these lists and consult the corresponding logical formalization. Third, these lists invite alternative formalizations of commonsense psychology, providing a starting point for those that want to pursue radically different approaches with identical conceptual breadth. Fourth, it serves

as a new lexical resource for future natural language processing systems, especially those that aim to interpret language using commonsense knowledge.

4.1 KNOWLEDGE MANAGEMENT

Knowledge Management concerns the basic reasoning and inference processes that lie at the core of the commonsense view of human psychology. The conceptual scope of this area, evident in the following linguistic examples, does not veer far from the philosophical concerns in the classical Platonic dialogue *Theaetetus*, which put forth a definition of knowledge as justified true belief, and the field of epistemology that follows from this work. Also included are process-oriented concepts within this representational area, such as the mental actions of adding and removing beliefs and assumptions, checking for inferences, the realization of consequences, and the affirmation of beliefs.

1. Managing knowledge ability: smart, gifted, clever, bright, intelligent, sharp, witty, brilliant, brainy, astute, quick learner, logical, rational, deluded, delusional, retarded, mentally handicapped, not the sharpest knife in the drawer, not the sharpest tool in the shed
2. Bias toward belief: gullible, believer, credulous, tend to believe, easy to convince
3. Bias toward disbelief: doubtful, skeptical, incredulous, naysayer, doubter, not born yesterday
4. True: correct, right, proper, fact, truth, reality, veracity, honest, accurate, verisimilitude, is the case, truly, valid
5. False: wrong, mistaken, incorrect, in error, inaccurate, amiss, awry, falsehood, falsity, fictitious, invalid
6. Truth value: whether or not it is true, true or false, validity, authenticity, credibility
7. Proposition: statement, sentence, position, idea that
8. Belief: think that, believe that, figure that, imagine that, reckon that, opinion, impression, feeling that, hypothesis, sense that, understanding that, estimation
9. Unknown: ambiguous, not known, unsure, uncertain
10. Revealed incorrect belief: mistakenly thought, used to believe that
11. Assumption: presumption, presupposition, premise, supposition
12. Unjustified proposition: hunch, guess, superstition
13. Partially justified proposition: to the best of one's knowledge, as far as one knows, reasonable, not impossible, possible, plausible
14. Justified proposition: logically necessary, logically follows that, no possible objection, justified, proof, fundamental, grounds for, owing to, on account of, validated by, beyond doubt, without a doubt, sure, certain
15. Inference: generalization, inferential, speculative, conjectural, conjecture, judgment
16. Consequence: conclusion, implies that, implication, corollary
17. Justification: rationale, theory behind, pretext for, foundation for, warrant, evidence that
18. Poor justification: does not follow from, not logical, flawed proof
19. Circular justification: circular argument, specious argument, circular reasoning

20. Contradiction: inconsistent, paradox, incongruity, conflicting beliefs, discrepancy, oxymoron, contrary notions
21. Knowledge: info, information, data
22. Knowledge domain: area of knowledge, area of specialization, area of expertise, field of study
23. Partial domain knowledge: some knowledge of, dabbles in, somewhat familiar with, working knowledge of
24. World knowledge: matter of fact, first-hand knowledge, second-hand knowledge, experiential knowledge, experience, hands-on knowledge
25. World model knowledge: theory of, science of, paradigm, world view, way the world works
26. Shared knowledge: commonsense, common thought, widespread belief, general conception, popular opinion, everyone knows that, everyday notion, commonly accepted that
27. Add belief: learn, find out, ascertain, discover, determine, apprehend, accept that, start believing, start to think that, begin to feel that, form the opinion, come to terms with, come around to one's point of view
28. Remove belief: change one's mind, lose faith in, stop thinking that, quit believing in, become unsure
29. Add assumption: assume that, suspect that, postulate, guess that, suppose that
30. Remove assumption: stop counting on, quit assuming, get back to reality
31. Check inferences: consider, think it over, draw inferences, judge, reckon, think about it, reason
32. Ignore inference: never mind that, putting aside that
33. Suppress inferences: without considering, disregarding the fact that, try not to think about, refuse to consider
34. Realize contradiction: conclude a contradiction, realize an inconsistency
35. Realize: conclude, become aware, occur to, dawn on, figure out
36. Knowledge management failure: confused, baffled, puzzled, bewildered, perplexed, mystified, dumbfounded, head in the clouds, lost
37. Reaffirm belief: verify, corroborate, confirm, affirm, reaffirm, make sure that

4.2 SIMILARITY COMPARISONS

Similarity Comparisons concerns the comparison of things, real or imagined, based on their representations. Along with basic notions of similarity and difference, this area includes other mental processes that are based on similarity judgments, specifically the ability to recognize patterns and draw analogies between otherwise dissimilar things.

1. Similarity comparison ability: able to differentiate, good at seeing the differences, be discerning
2. Compared things: comparison set, greater than, more than
3. Make comparison: compare, differentiate, distinguish, tease out the differences, equate, liken
4. Draw analogy: abstract comparison, see a connection, find a similarity
5. Find pattern: like last time, recognize a pattern, see a sequence, notice a trend

6. Comparison metric: dimension by which, variable of interest, distinguishing feature, basis for comparison, range of comparison
7. Same characteristic: have in common, share the feature, common characteristic, similar trait, both share, both have, grouped according to
8. Different characteristic: distinguished by their, both differ in, contrasting feature, different in that, differing in their, same except for, separable by
9. Analogical mapping: corresponds to, maps onto, aligned with, parallels, in the analogy
10. Pattern: trend, tendency, inclination, motif, order
11. Similar: resembles, takes after, has a resemblance to, bears a likeness to, shares features, has much in common, spitting image, chip off the old block, two of a kind, nearly identical, alike, comparable, no difference between
12. Dissimilar: different, contrasting, antithetic, differing, opposite, separable, not the same, bear no resemblance, nothing in common, does not take after
13. Analogous: simile, metaphoric, share an abstraction, the same in the abstract

4.3 MEMORY

In the commonsense view, memory is a metaphorical container for thoughts. Storing and retrieving from memory involve the transfer of thoughts in and out of this store to one's focus of attention. Included are concepts involved in deliberative memory retrieval as well as for passive retrieval, where a memory cue triggers an involuntary reminding.

1. Memory ability: knowledgeable, wise, well-informed, know by heart, know word for word, never forget, have a great memory
2. Memory item: memory, nostalgia, remembrance, recollection, past knowledge, experience, memoir, wisdom, memorable, unforgettable
3. Repressed memory item: repressed memory
4. Memory storage: cause to remember, learn, learn by heart, memorize, learn word for word, commit to memory
5. Memory retrieval: remember, think back, recall, recollect, reminisce, summon up, think of, retain, bear in mind, rings a bell, relive, fresh in one's mind, stick in one's mind, commemorate
6. Memory retrieval failure: no memory of, does not come to mind, lose the memory of, slips the mind, draw a blank, forget, tip of the tongue, rack one's brain
7. Memory repression: repress a memory, bury the thought, clear the thought, purge your mind, forget it
8. Reminding: reminiscent of, evocative, remindful, triggers a memory, reminds one of
9. Memory cue: souvenir, memento, keepsake, memorandum, reminder, cue, hint, help remember
10. Schedule plan retrieval: will not forget to, mental note to, remind one to, remember to, be sure to
11. Scheduled plan retrieval: remembered to, did not forget to
12. Scheduled plan retrieval failure: forget to, did not remember to, slipped one's mind

4.4 ENVISIONING

What are you thinking about? It is a common question that people can easily answer, and one that we will want to pose to robots in the future. Just as with people, we expect that robots should respond with one of handful of types of reports. They might be thinking about the past and how things got to be the way they are now. They might be focused on the present moment and the current state of the world. They might be wondering about the future and what is going to happen next. Perhaps instead they are daydreaming, imagining fictional places and people that have no temporal or causal connection to the world that they are experiencing. These processes all have a few important commonalities. They each involve the manipulation of conceptual models of the world, real or fictional. They each are concerned with events and states within these models that have a temporal and/or spatial extent, and the temporal and causal relationships between them. Each involves the consideration of the possible, freely considering counterfactual events and states as easily as those that are experienced in reality.

The phrase “thinking about” is probably the most commonsense way of referring to the suite of reasoning processes that share these commonalities, but in this book we instead use the term “envisioning.” The concepts in this area characterize the infrastructure of causal reasoning, including branching causal timelines of various likelihoods that organize states and events into a system of relationships. Included as well in this area are concepts that pertain to the management of a person’s current model of the world, such as advancing this model through the current moment, sending the present to the past and the future to the present, and validating or invalidating the expected model.

1. Envisionment ability: rich imagination, capacity to imagine, capability of foreseeing, imaginative
2. Envisionment failure: have trouble imagining, unimaginable
3. Envisionment preconditions: unforeseeable, before one can think about
4. Envisionment constraint: try not to think about, avoid imagining, keep oneself from thinking
5. Real-world envisionment: actually may happen, really has occurred, in fact will take place, in the real world, actually transpired, nonfiction, in reality
6. Other-world envisionment: fictional, dream, fantasy, fable, fantastical, not reality, only in one’s imagination, not really happen, not actually occur
7. Consistent envisionment: both could happen, consistent with the fact that
8. Inconsistent envisionment: cannot both be the case, inconsistent with the fact that
9. Envisioned state: how one imagines it, one’s vision of
10. Envisioned current world state: the way the world is, the way things are, present state of things, current view of the world, how one imagines the world to be, current state of affairs, at the current time, state of things now
11. Envisioned past state: view of history, way one thought it was, imagined past, knowledge about what happened
12. Envisioned future state: view of the future, way one thinks it will be, imagined future, beliefs about what will happen
13. Causal relationship: causal connection, causally related to, direct connection between

14. Causal influence: has something to do with, tends to cause, influences, play a role in
15. Causal chain: chain of events, string of events, cascading states, series of causes, domino effect
16. Causal system: scope of feasibility, realm of possibility, conceivable circumstances, network of influences
17. Propagated effects: aftermath, aftereffects, repercussions, in the wake of, consequences of
18. Envisionment branch point: hinges on whether, determining factor
19. Envisionment branch: possible outcome, potential consequence, contingency, fate, potentiality
20. Envisioned likelihood: the odds of, the probability that, the chances of
21. Definite likelihood: inevitable, will definitely happen, in all likelihood
22. Zero likelihood: impossible, will definitely not happen, an impossibility
23. Greater likelihood: more probably, higher chances of
24. Greatest likelihood: the odds are that, most likely
25. Random chance: luck of the draw, pure luck, due to chance, randomness, by hazard
26. Possible causal chain: conceivable fashion, always a way
27. Impossible consequence: no way that, could not have occurred
28. Combined envisioned likelihood: overall odds, joint probability, total chance that
29. Imagine state: imagine, envision, visualize, picture in one's head, once upon a time, in the mind's eye
30. Merge envisionments: dream became reality, fiction turned real
31. Advance envisioned current world state: the moment has arrived, the time is now
32. Correct envisionment: accurately imagined, just as one had imagined
33. Incorrect envisionment: mistaken vision of, had imagined it differently
34. Validate envisioned state: confirmed that, ensured that, checked that
35. Invalidate envisioned state: did not turn out that way

4.5 EXPLANATION

People want to know why. How did the world end up this way? What caused this to happen? The answers to these questions are explanations, the product of the mental process of explanation. Explanation is the side of envisionment that extends models of interconnected events backwards in time, to what came before to cause other events that are considered. The concepts below outline a multistage explanation process that begins with an attempt to explain a mystery, and is followed by the generation and evaluation of candidate explanations that are either adopted or rejected.

1. Explanation ability: critical thinker, deductive ability
2. Cause: induce, give rise to, bring about, lead to, determine, invoke, force, spawn, make happen
3. Mystery: enigma, riddle, conundrum, puzzle, unknown reason, no known cause

4. Explanation criteria: necessary to account for, needed to explain
5. Explanation: be because, owing to, as a result of, due to, by virtue of, on account of, set of factors, be why
6. Candidate explanation: a variety of reasons, another explanation, or because of, perhaps it was because
7. Best candidate explanation: the true reason, the real explanation
8. Factor: part of the reason, be causally involved, due to a combination of
9. Explain: figure out why, imagine why, conceive of why, to account for, elucidate why, come up with an explanation
10. Attempt to explain: consider why, think about why
11. Generate candidate explanations: brainstorm for why, hypothesize that, speculate that
12. Assess candidate explanations: scrutinize the hypothesis, consider the theory that
13. Adopt explanation: accept that the reason, adopt the theory that
14. Explanation failure: for lack of a better explanation, failure to account for, not understanding why
15. Explanation generation failure: no conceivable reason why, heaven only knows why
16. Explanation satisfaction failure: no reasonable explanation, no acceptable reason why
17. Unsatisfying explanation: poor explanation, rejected account of why
18. Explanation preference: prefer explanations that, look for explanations that

4.6 MANAGING EXPECTATIONS

What is going to happen next? Expectations are envisionments of causal consequences, specifically those that are imagined to follow in the future from the current world state. Expectation is the side of envisionment that extends interconnected models of states and events forward in time, toward the temporal and causal consequences of other imagined states. As the world reveals itself, these expectations may be confirmed or violated, the latter of which is the basis for the state of surprise.

1. Expected event: expectation, prediction, anticipated event, be expected
2. Unexpected event: surprise, a twist, out of nowhere, a shock, be unforeseen, fluke, the unimaginable happened
3. Expectation justification: reason for expecting, basis for prediction
4. Add expectation: predict, forebode, foresee, expect, destine, anticipate, envision what happens next
5. Remove expectation: no longer expecting, give up hope that, stop waiting for
6. Expectation ability: prophetic, oracular, visionary, seer, clairvoyant
7. Expectation confirmation: nothing out of the ordinary, just as one had expected, satisfy one's expectations
8. Expectation violation: surprised, astonished, amazed, astounded, dumbfounded, flabbergasted, startled, stunned, taken aback, taken by surprise, shocked, blindsided

4.7 OTHER-AGENT REASONING

Other-Agent Reasoning concerns the imagining of the mental states of other people, as well as introspectively considering one's own mental state. By including these concepts here, we are arguing that a theory of thinking about mental states is itself part of commonsense psychology, that is, that how people think they think includes beliefs about how people think about how people think! This idea is perhaps best understood in consideration of the representational concept of a psychological theory, included below, for which the theory formalized in this book is an instance. Despite the title of this representational area, it is also meant to include the mental processes of self-reflection and introspection, where people think about their own mental processes from the privileged perspective.

1. Other-agent reasoning ability: empathetic, telepathic, inconsiderate, self-centered
2. Psychological theory: psychology, the way people think, theory of mind, way the mind works
3. Other-agent model: know about someone, familiar with someone, acquainted with someone
4. Other-agent envisionment: the way one sees it, from one's perspective, through one's eyes, put oneself in someone's shoes, read someone like a book, see where one is coming from, what is going on in one's head
5. Other-agent envisionment failure: not understand why one would, have the element of surprise
6. Introspection: look inside one's own thoughts, ask oneself what one
7. Introspection failure: not know what I was thinking, surprised oneself

4.8 GOALS

People want a lot of things: a loving partner, a fulfilling career, parents who are well cared for, an attractive physique, a creative outlet, and material possessions. These are the sorts of goals that people target in their pursuit of happiness. The central importance of goals in people's lives is reflected in the richness of language about goals. The list of goal concepts below are abstract goal categories, rather than specific goals that people choose to pursue. A specific instance of a goal may change its class over time. For example, the preservation goal of caring for the health of one's parents may first only be an envisioned future goal when they are young and in good health, a shared goal among siblings when their health needs attention, an auxiliary goal when they are involved in one's plans in some way, and a violated preservation goal when their health turns bad.

1. Goal: ambition, intention, hope, desire, objective, purpose, aim, aspiration, calling, yearning, hunger for, craving, thirst for, aching for, pining for, long for, wish
2. Subgoal: toward the larger goal, waypoint, step along the way to
3. Conflicting goal: impossible to have both, not have one's cake and eat it too
4. Auxiliary goal: be nice as well, also do with, great if also, also if at all possible
5. Shared goal: common aim, shared objective, everybody wants
6. Collaborative goal: goal of the group, team objective, aim of the organization
7. Competitive goal: rivalry for, competition for, compete for, vie for

8. Adversarial goal: fight for, duel for, war against, battle against, thwart, undermine, foil
9. Unknown goal: mysterious objective, unspecified aim
10. Nonconscious goal: sure one wants, know one's own desires, subconscious wish, unconscious desire
11. Achieved goal: get what one wants, accomplishment, victory, achievement
12. Unachieved goal: yet to achieve, still to accomplish, unfulfilled desire, elusive
13. Unachievable goal: pipe dream, vain hope, unrealistic goal
14. Preservation goal: goal to protect, goal to preserve, aim to maintain, aim to uphold
15. Violated preservation goal: failure to protect, failure to preserve, failure to maintain
16. Persistent goal: endeavor, crusade, the movement toward, unending fight against, eternal struggle
17. Envisioned future goal: one day will want, not yet want
18. Envisionment goal: want to imagine, want to dream
19. Planning goal: want to know how to
20. Execution goal: want to do, want to get it done
21. Knowledge goal: desire to understand, curious, curiosity, inquisitive, nosiness

4.9 GOAL THEMES

To capture a commonsense intuition, Schank and Abelson (1977) introduce the term “goal theme,” which “contains the background information upon which we base our predictions that an individual will have a certain goal.” In their formulation, the attribution of a given goal theme to a person provides a causal account for other goals that would otherwise be senseless if viewed in isolation. Three broad classes of goal themes were described by Schank and Abelson, which can be referred to as role, relationship, and life themes. We include the concept “thrive,” the causal antecedent of all of a person's goals.

1. Goal theme: source of one's desires, gives rise to the goal, ought to want, is supposed to want
2. Role theme: want to be a good, one's job to want
3. Relationship theme: want to do right by, for the sake of the relationship
4. Life theme: life's ambition, purpose in life, calling in life
5. Thrive: pursuit of happiness, ultimate purpose, prosper, flourish, make it in this world

4.10 THREATS AND THREAT DETECTION

The commonsense concept of a threat is somewhat remarkable in that it sits at the overlap between so many different ideas in commonsense psychology, particularly those related to one's goals, one's plans, and expectations about the way the world is going to be. Combined, these ideas provide a concise definition: a threat is an expectation that if something is not done, there is some chance that some goals will not be achieved. In everyday English, the term also often refers to the causal antecedents

of the expected world state, particularly where these causes are agents, for example, “The rogue state is a threat to world peace.”

1. Threat: danger, risk, hazard, ominous, rocky road, thin ice, dangerous ground
2. Goal in threat: be threatened, a threat to, be put in harms way
3. Threat condition: threat hinges on, danger is caused by, ripe for disaster
4. Threat effects: fallout, aftermath, post-apocalyptic, unfortunate consequences, bad outcomes
5. Possible threat: serious threat, prospective danger, not foolproof, danger is real, imminent threat
6. Impossible threat: safe bet, worry over nothing, no fear, nothing to be concerned about
7. Realized threat: not pull it off, fears came true, luck pressed too far, be screwed, not catch a break
8. Unrealized threat: pulled it off, caught a break, came through unscathed, slipped under the radar, dodged a bullet, close call, close one, false alarm, narrow escape, empty threat
9. Threat detection: think of everything, look for chinks in the armor, look before you leap, assessed the risks, considered the pitfalls, thought about the dangers
10. Add threat: raise a red flag, worry about, be concerned with, see the danger, afraid that
11. Remove threat: stop worrying, fears lifted, become unconcerned
12. Threat detection failure: caught off guard, wake up call, caught with one’s pants down, sucker punch, not read the writing on the wall, fail to consider the risks
13. Threat assessment: think it is worth it, worth the risk, worth the gamble, venture, calculate the risks
14. Threat detection ability: can judge the risks, can play the odds
15. Bias against risk: paranoid, overly nervous, scaredy-cat, coward, worrywart, cautious, careful, conservative, risk averse
16. Bias toward risk: daring, audacious, adventurous, gambler, reckless, throw caution to the wind

4.11 PLANS

Here the term “plan” is used in the same sense as in this quote from Daniel Burnham (1864–1912), the celebrated American architect and city planner, “Make no little plans. They have no magic to stir men’s blood and probably will not themselves be realized” (Moore, 1921). Burnham’s quote is emblematic of his design philosophy, as evidenced by his development of the influential Burnham Plan for the city of Chicago, but the quote also directly implies some of the characteristics of plans in general. First, plans are seen as something that are made by people, as Burnham advises the listener to do. Although Burnham’s plans often were presented as physical objects (architectural drawings), it is natural as well to view plans as entirely mental constructs, the product of human reason. In the commonsense view, therefore, a plan is a thing that resides in one’s head. Second, plans can be small or large. For Burnham, the size of a plan may have been directly correlated to the size of the

building, neighborhood, or city that he was envisioning, but a plan's size may also be based on the quantity or severity of the steps involved, or simply the duration that it would take to execute. Third, one person's plan can be communicated to other people for appraisal. When Burnham says that small plans cannot "stir men's blood" he is noting that people can have the plans of other people in their head to appraise their value, without the necessity that those plans be adopted or pursued. The idea that someone could have in his or her head an "un-adopted plan" makes perfect sense in everyday discourse, but much of the theoretical work in artificial intelligence (AI) planning seems to struggle uncomfortably with this notion, leading to a terminology of desires, intentions and commitments that take on nonobvious meanings within the literature. Fourth, plans are either realized in the world or not. For Burnham's realized plans, the world is changed in readily apparent ways: old roads and buildings are demolished, new ones take their place. Other plans may leave no obvious trace of their execution, such as a plan to have a casual conversation with a friend, or to relax at home on a Saturday afternoon. Despite the lack of obvious change to the world, the realization of these plans is dependent on the occurrence of real-world events.

1. Plan: program, scheme, project, plan of action, plot, roadmap, blueprint, agenda, proposal, suggestion, course of action, approach
2. Partial plan: underspecified plan, indeterminate plan, sketchy plan, incomplete plan, half-baked
3. Strategy: tactic, stratagem, abstract plan
4. Strategy instance: example of the strategy, instantiation of the strategy
5. Reusable plan: familiar plan, old trick, procedure, recipe, technique
6. Normal plan: normal thing to do, usual course of action, conventional action, classic approach
7. Adversarial plan: plan of attack, battle plan, diabolical plan
8. Counterplan: plan of defense, counter proposal, defensive stance
9. Assistive plan: plan of assistance, plan of support, plan in support of, aid program
10. Competitive plan: playbook, play, game plan
11. Shared plan: common plan, everyone doing the same thing, everyone had the same idea
12. Collaborative plan: plan to cooperate, plan to work together, partnership, collaboration
13. Envisioned plan: think what one will do, imagine one's plan, one's conceivable plan
14. Unknown plan: not know what the plan is, mysterious plan, the divine plan
15. Nonconscious plan: subconscious plan, unconscious plan, tacit knowledge of how to, force of habit
16. Continuous plan: never-ending plan, endless plan of action, plan to continue indefinitely
17. Periodic plan: routine, regiment, ritual, plan to periodically
18. Repetitive plan: repetitive course of action, plan to repeat
19. Occasional plan: plan to occasionally, plan to sporadically, plan to sometimes
20. Executed plan: completed plan, performed plan, what one did, finished doing, executed as planned

4.12 GOAL MANAGEMENT

Goal Management concerns the mental processes involved in managing the lifecycle of one's goals. Goals are seen as entities that can be adopted, suspended, resumed, abandoned, violated, or achieved. A further distinction is made to identify adopted goals that are currently being pursued, that is, the focus of one's planning or execution capacities. People are also viewed as having multiple goals at any given time, with prioritization among them that may be influenced through deliberative assessments.

1. Goal management ability: in touch with one's needs, have one's priorities straight
2. Add goal: resolve to, make up one's mind to, take up the goal of, now want
3. Abandon goal: stop wanting, throw in the towel, call it quits, lose hope, not want anymore
4. Suspend goal: set aside the goal, shelf the desire, put one's goals aside
5. Achieve goal: achieve, accomplish, succeed in, obtain the goal of, cross off one's list
6. Violate goal: be crushed, be vanquished, be foiled, kill one's dreams
7. Modify goal: settle for, try for a different goal, lower the bar, rethink what one wants
8. Specify goal: clarify one's goals, qualify what one wants
9. Generalize goal: pursue a broader goal, broaden one's hopes, be less picky
10. Goal justification: reason for wanting, why one wants, explanation for the desire, underlying motivation
11. Goal assessment: think about what one wants, ask oneself if it is worth it, reevaluate one's intention, rethink one's aim
12. Resolve conflicting goals: decide among goals, sort out contrasting aims
13. Goal preference: most important, more desirable, rather have, better to want
14. Prioritize goals: set one's priorities, prioritize one's goals, order one's intentions
15. Goal set: wish list, set of desires, things that one wants
16. Currently pursued goal: current ambition, latest desire, present aim, what one wants now

4.13 EXECUTION ENVIRONMENT

One of the main purposes of envisioning is to imagine how action could change the future, or explain the past. Concepts of Execution Environment are fundamental across all envisioning processes, for example, explanation and expectation, but are distinguished in that each of them involves the envisioning of the causal implications of some action of an agent, whether they be one's own or that of others. These concepts form the conceptual foundation for planning one's own actions to achieve goals.

1. Envision execution: imagine doing, rehearse in one's mind, see oneself doing
2. Execution environment failure: not imagine how one, not visualize the way one
3. Agency: acting, to act, deed, doing something, a behavior, the act of
4. Side effect: secondary effect, collateral damage, unintentional consequence
5. Opportunity: take advantage of, serendipitously, see an opening, capitalize on

6. Limit of failure: hanging on a thread, breaking point, push one's luck
7. Moment in execution: in the midst of, in the course of
8. Arbitrary moment in execution: any moment while doing, throughout the course of doing
9. Planner: see oneself doing, picture oneself
10. Envisioned execution duration: time needed to, the time it takes to, manhours
11. Envisioned likelihood of success: sure thing, slam dunk, good shot at, odds of succeeding
12. Envisioned failure: appears hopeless, lost cause
13. Arbitrary execution context: could be done anywhere, works in any circumstance, applicable in any situation

4.14 CAUSES OF FAILURE

Try as we might, failure is common. Attributing blame to the root cause of failure is a central challenge in explanation. To simplify our efforts, we often resort to patterns of plan execution explanations that are widely applicable to everyday situations.

1. Poor plan: ignorance, inexperience, naivety, stupidity
2. Poor conditions: perfect storm, bad luck, adverse circumstances
3. Poor execution: ineptitude, impotence, inability, inefficacy

4.15 PLAN ELEMENTS

Plan Elements are those concepts that further elaborate the internal structure of plans, enumerating their component parts. In addition to the concept of a subplan and its varieties, this area includes all of the programmatic control structure that defines a policy for how plans should be executed, including branch points, iterations, repetitions, and preconditions.

1. Subplan: milestone, step, part of the plan, stage in the plan, small part of the plan
2. Partial subplan: step is sketchy, underspecified part of the plan
3. Subplan agency: who does a part, delegated to, part executed by
4. Planned termination: final part of the plan, moment it is finished, close of the plan, after the last step
5. Subplan ordering: series of steps, sequence of actions, ordered steps, chain of actions
6. Precondition: need to first, in order to, before one does
7. Knowledge precondition: need to know first, not do without knowing, necessary knowledge, information needed
8. Resource precondition: need to have first, not do without having, necessary resource, resource needed
9. Conditional branch point: point of divergence, variability in the plan, point of contingency
10. Conditional branch: contingency, option
11. Branch point conditions: at which point it depends on
12. Else conditional branch: the other option, otherwise one plans to, Plan B

13. Termination conditional branch: in that case one is done, otherwise one is finished
14. Iterative subplan: do the same to each, repeat for every
15. Planned iteration quantity: times one planned to
16. Iteration quantity: number of ones one has done
17. Iteration termination condition: reason for leaving ones undone, case one should stop before completing all
18. Repetitive subplan: step over and over, repetitive part of the plan
19. Repetition termination condition: repeat until, keep at it until, do it as long as
20. Repetition quantity: number of times repeated, repetition count
21. Periodic subplan: step repeated every, repeat at regular intervals
22. Periodic subplan period: time between repeating the step, interval between repetitions
23. Reactive subplan: to do in case
24. Reactive subplan condition: while being on the look out for
25. Plan duration: do over a period of, time allotted for
26. Required precondition: need regardless, either way one needs, in any case it is required
27. Required subplan: need to do regardless, either way one needs to, has to be done in any case

4.16 PLANNING MODALITIES

Planning Modalities enumerate the different styles of reasoning that people employ in decided what to do, broadly speaking. The most general concept is that of goal pursuit, an abstraction that includes deliberative plan construction, the adaptation of previous plans, the collaborative negotiation of plans with others, and several other planning modalities. Concepts related to three of these modalities, namely plan construction, plan adaptation, and design, are further elaborated in subsequent representational areas.

1. Goal pursuit: working to achieve, making progress on the goal, working toward
2. Planning: plan to, think of a plan, draft a plan, formulate a plan, map out a plan, lay out a plan, make a plan, devise a plan
3. Planning ability: schemer, crafty, cunning, able to plan
4. Adversarial planning: conspiracy, cabal, devise a battle plan, plan to attack
5. Counterplanning: plan a defense, plan to defend, devise a counterplan
6. Assistive planning: figure out how to help, plan to support, devise a plan to aid
7. Collaborative planning: put our heads together, plan together
8. Competitive planning: devise a game plan, straight out of the playbook
9. Plan construction: figure out the steps, assemble a plan, pull together the parts of the plan
10. Plan adaptation: adapt a plan, salvage a plan, reuse a plan, apply a plan to a situation
11. Modify currently executing plan: change directions, change one's ways, break from the plan
12. Scheduled plan modification: last-minute changes, change at the last minute

13. Single-goal planning: one-track mind, single minded, stay focused on a goal, planning solely to
14. Multiple-goal planning: kill two birds with one stone, plan to get as much done as possible
15. Designing: come up with a design, work out a design, sketch out a design
16. Other-world planning: in that situation one would, if things were different one would
17. Other-agent planning: recommendation, if I were you I would, in your situation I would

4.17 PLANNING GOALS

Planning Goals concerns the goals that a person is trying to achieve in the process of deciding upon a plan of action. Here we are making a distinction between the types of goals that people are devising plans to achieve, and the preferences and priorities that people use to decide between candidate plans. Several of the concepts of planning goals concern the optimization of a resulting plan, such as the search for plans that take the least amount of time (minimize duration) or money (minimize value). Other planning goals prefer plans that include particular actions or events, or create opportunities or obstacles for other people. This area also includes three concepts for the goal of locating something (a person, place, or thing) that will play some other role in the plan.

1. Include action: try to incorporate the step, include in the plan if possible
2. Avoid action: only as last resort, try to avoid doing, streamline the plan
3. Enable event: set the stage for, set the conditions for
4. Block event: prevent against, safeguard against, take preventive measures
5. Enable threat: work toward one's demise, set one up to fail
6. Block threat: take preemptive action, alleviate the risk
7. Enable transfer: clear the path, remove obstacles, clear the way, open the flood-gate
8. Block transfer: secure in place, prevent from leaving, keep out, obstruct
9. Enable agency: make it possible to, enable one to
10. Block agency: imprison, paralyze, keep one on a short leash, have someone under one's thumb
11. Enable other-agent goal satisfaction: make someone happy, stay out of one's way
12. Block other-agent goal satisfaction: thwart, derail, undermine, hinder, get in one's way
13. Minimize value: less is more, frugal, thrifty, be conservative with, optimal amount, miserly
14. Maximize value: bigger is better, greed, go for broke, as much as possible, be liberal with
15. Minimize duration: time is of the essence, as fast as possible, not waste time, efficient
16. Maximize duration: prolong, stretch the moment, make it last forever, as long as possible
17. Locate thing: look for, search for, locate, get one's hand on

18. Locate agent: find someone to, search for the person who, fill the job, talent search
19. Locate location: survey, pinpoint, find the place where, scout locations
20. Maintain plan progress: avoid backpedaling, keep moving forward, not undo what one has accomplished

4.18 PLAN CONSTRUCTION

Among the various ways that people decide what to do, Plan Construction is the process of devising plans from scratch. The process involves reasoning about the causal consequences of one's actions, and assembling a plan that would theoretically achieve the goal if correctly executed. This style of planning most closely aligns with the classical planning paradigm in AI, where operators are selected and ordered through automated means–ends analysis.

1. Suspend planning: hold off on the details, put aside the plan
2. Resume planning: finish the plan, take up the plan again
3. Abandon planning: stop planning, give up on the plan
4. Specify plan: flesh out the details, be specific about the plan
5. Add subplan: add a step, addendum to the plan, put a part in the plan
6. Remove subplan: remove a step, simplify the plan
7. Identify precondition: realize the need to first, see a necessary step
8. Select subplan: choose a step, select an action
9. Order subplans: first things first, figure out the order
10. Obstacle: barrier, impediment, impasse
11. Planning problem: challenge, dilemma, sticky situation, something must be done
12. Planning option: one way to do it, have some leeway
13. Planning decision: decide what to do, choose the best approach
14. Resolved planning problem: get around the problem, solution to the dilemma
15. Planning preference: ideal plan, prefer a course of action that, favor plans that
16. Planning constraint: be constrained by, be under constraints, be limited by
17. Replan: start from scratch, back to the drawing board, try a different approach
18. Replan subplan: patch the plan, fix part of the plan, rework a step
19. Candidate plan: one way to, possible approach, reasonable course of action
20. Assess plan: evaluate the plan, scrutinize the plan, review the plan
21. Select candidate plan: choose the plan, pick the best plan
22. Selected candidate plan: plan one decided on, the original plan, best course of action
23. Unselected candidate plan: Plan B, plan one considered, rejected plan
24. Successful planning: wrap up the plan, finalize the plan, complete the plan
25. Planning failure: at a loss, uncertain what to do, not know what to do, stumped

4.19 PLAN ADAPTATION

Instead of dreaming up new plans to familiar problems, it is often easier just to do the same thing that worked last time. Things are not exactly as they were before, so it may be necessary to adapt to a changed set of circumstances. In the commonsense

view of Plan Adaptation, people retrieve old plans that they had for previous situations and modify them in various ways to make them applicable to the current conditions. Sometimes these changes involve the addition or subtraction of plan steps, the modification of magnitudes or parameters, or the assignment of roles in multiperson plans. Some plans are straightforward to adapt, while others take a lot of mental work. In the end, people end up with plans that they like or plans that they reject.

1. Retrieve plan: remember how to, consider an approach, recall a technique, pull from the playbook
2. Modify plan: apply to this situation, adapt to this case, tweak to make it work, rethink the approach
3. Modify plan element: tweak a step, swap an action, substitute a step, amend a part of a plan, change it up a bit
4. Modify plan value: use more, do it with more, try increasing, correct for
5. Modify plan agency: stand in for, play the role of, fill in for, take over for
6. Adaptation cost: difficult to apply, complex application of, straightforward application
7. Successful adaptation: successful application of, thoroughly reworked the plan
8. Failed adaptation: not applicable here, unsalvageable plan, not serve one's purpose

4.20 DESIGN

Design is the planning of a physical configuration, of information to be communicated, or of an object intended to serve some function. The product of this reasoning need not include any actions to be executed, as when we talk about the successful completion of an architectural plan (e.g., the Burnham Plan for the city of Chicago). Instead, the goal is a satisfactory or optimal configuration of some composite entity that adheres to design constraints while supporting an intended use.

1. Design: layout, configuration, invention, model, schematic, arrangement
2. Design constraint: boundary, within limits, design restriction, up to code, limitations, specifications, requirements
3. Intended use: primary use, standard usage, expected application, supposed to be used, function
4. Unintended use: repurposed, nonstandard usage, jury-rigged, unusual application
5. Flawed design: defective design, design failure, engineering failure, bad design, hack
6. Design adherence: precision construction, faithful to the design, fidelity, stick to the design, meet specifications, according to design
7. Generate design: think about the design, working out the layout, imagining the structure, configure, arrange
8. Adapt design: customize the design, modify the design, design inspired by
9. Design failure: unsatisfying design, impossible layout, inconsistent configuration

4.21 DECISIONS

A decision is the selection of a choice among options. The concepts listed in this area take the perspective that decisions should be viewed as a particular class of plan element, usually a part of a larger plan; for example, you'll decide which way to go when we get there, rather than as a distinct planning modality. The process of decision making is compositional, involving the enumeration of a set of choices, deliberation over these choices, selection of the best choice, and (possibly) revision of one's choice after the fact.

1. Decision: decision to make, choice to make, runoff, election
2. Enumerate choice set: identify options, list choices, think of the possibilities
3. Deliberate: ruminate, ponder the choices, mull over one's options, think about the possibilities
4. Selection: make a choice, settle on, go with, pick, decide on, endorse, vote on
5. Decision revision: change one's mind, fickle, renege on, to waffle, vacillate, go back on
6. Suspend decision: sleep on it, leave it on the table, keep one's options open, put off the decision
7. Resume decision: take up the decision, finish deciding
8. Decision series: set of choices, chain of decisions, series of selections
9. Choice set: possible choices, set of options, range of candidates
10. Possible choice: in the running, candidate, prospect, alternative, option
11. Choice characteristic: determining feature, distinguishing characteristic, pros and cons
12. Best choice: frontrunner, favored option, preferred choice, cream of the crop, leading candidate
13. Selected choice: one's pick, one's choice, the one chosen, one's decision
14. Unselected choice: not make the cut, decided against, runner up, rejected choice, not chosen
15. Previously selected choice: the usual, consistent choice, ordinary choice, previous pick
16. Envision choice consequences: consequences of choosing, look before you leap
17. Cost-benefit analysis: evaluate the pros and cons, weight the decision
18. Choice criteria: evaluation metric, yardstick, benchmark, litmus test
19. Choice preference: prefer candidates that, preferential treatment, affirmative action
20. Obvious decision: no brainer, clear choice, clear cut, choice is evident, no deliberation needed
21. Insignificant decision: unimportant decision, inconsequential, immaterial, choice is irrelevant
22. Decision justification: reason for choosing, selection based on, motivation for picking
23. Decision factor: deciding factor, influenced one's decision, play a role in the decision, part of the decision
24. Arbitrary decision justification: random choice, on a whim, impulsive decision, blind choice, crap shoot, luck of the draw
25. Decision consequence: effects of the decision, live with one's choice

26. Positive decision consequence: good choice, excellent pick, decision bears fruit
27. Negative decision consequence: disastrous decision, poor choice, mistake

4.22 SCHEDULING

Scheduling sits at the border between planning and execution, between dreaming and doing, between intention and action. The central concept in this area is that of a schedule, a mental agenda of the plans that one expects be executed at future points in time. The process of scheduling, consequently, is a deliberation concerning the time a plan should be executed.

1. Schedule: one's agenda, to-do list, on one's plate, things that need to be done, stuff to do
2. Schedule capacity: a full plate, full schedule, booked, overbooked, overloaded, busy, stretched thin, packed schedule
3. Unscheduled duration: free time, down time, spare time, nothing to do, idle time
4. Next unscheduled duration: have a moment, have a chance, have a second, have some time, next opening, soonest opportunity
5. Pending plan: wanting to do, wanting to find time for, intend to squeeze into one's schedule
6. Scheduled plan: be slated for, on the schedule, committed to do, appointment
7. Next scheduled plan: up next, next on the agenda, next order of business
8. Scheduled plan duration: time allotment, allotted time, period of performance
9. Scheduled start time: kick off, commencement, will start at, scheduled to begin
10. Scheduled end time: target time, target date, will finish at, scheduled to end, wrap-up time
11. Deadline: time limit, curfew, drop dead date, last possible moment, must be finished by
12. Start time deadline: before it is too late, must start before, strike while the iron is hot
13. Scheduled close to deadline: the eleventh hour, crunch time, down to the wire, time pressure, without a moment to spare, not a moment too soon, by the skin of one's teeth, in the nick of time
14. Scheduled far from deadline: time to spare, breathing room, plenty of time, time to kill
15. Schedule preference: expedite, rush through, rush job, pick up the pace, move it into high gear, as soon as possible
16. Schedule plan: make a date, make an appointment, put on the schedule, fit into the schedule, make time for
17. Schedule periodic plan: set a usual time, schedule regularly, schedule periodically
18. Unscheduled plan: postpone, some other time, put on the back burner, delay, put off, hold off
19. Reschedule plan: reschedule, push back until, push forward to, move up in the schedule
20. Indefinitely postpone: hold off forever, put off one's whole life, do in one's next life

21. Modify schedule: rework the schedule, throw out the agenda, free up some time, rearrange one's schedule
22. Collaborative scheduling: coordinate schedules, arrange schedules, choreograph
23. Scheduling failure: a mix up, double booked, not find time to, dropped the ball

4.23 MONITORING

Plans often include the intentional triggering of some action based on some conditions of the situation. We imagine that stockbrokers would monitor the market for conditions that would prompt them to sell their portfolio. High school students might monitor the wall clock when taking a test to adjust the amount of time that they allot to answering each test question. People driving on an interstate highway might monitor the fuel gauge, the distance to exits, and their own bodies in making decisions about when to stop for gas at a service station. In the list that follows the term "monitor" is used both as a verb referring to the mental event of considering whether a situation matches some trigger conditions, and as a noun referring to a type of mental structure that describes these conditions and the actions that are to be triggered. This latter meaning of monitoring suggests that a stack or set of monitors might be active at any given time for a person, and that the adding and removing of monitors from this stack would be part of any moderately complex plan. It is also within this representational area that we can characterize the concept of waiting, both waiting for and waiting to, as a type of monitor whose conditions trigger the continuation of the remainder of a plan.

1. Monitor: be observant, on the lookout, keep one's eyes peeled, waiting for, await, patrol, check, watchful, vigilant, observant, keep one's eyes open, keep tabs on, guard
2. Monitor thing: not take one's eyes off it, not lose sight of, eyeballing
3. Monitor agent: stakeout, breath down one's back, look over one's shoulder, spy on
4. Monitor self: observe oneself, self conscious, self restraint
5. Terminate monitor: stop waiting, be at ease, let down one's guard
6. Monitor trigger conditions: catch in the act, at first sign of, what one is looking out for
7. Monitoring frequency: how often one checks, time between checking
8. Duration of monitoring: on one's watch, while on patrol
9. Untriggered monitor: detect nothing, nothing alarming
10. Trigger monitor: be alarmed, sound the alarm, see what one is after
11. Monitoring failure: overlook, slip through the cracks, lose sight of, let one's guard down, catch off guard, be right under one's nose

4.24 EXECUTION MODALITIES

There are several ways that people go about doing things. Different execution modalities are distinguished largely by the degree to which the action was planned ahead of time or with respect to temporal factors. In some cases, these modalities are

necessarily exclusive, for example, an execution is unlikely to be both planned and spontaneous. In other cases, these modalities simply denote special concerns that an execution system would need to address, that is, the coordination needed for collaborative execution between multiple people.

1. Execution: the doing of, the act of
2. Planned execution: as intended, go according to plan
3. Spontaneous execution: extemporaneous, improvisation, make up as one goes along, wing it, by the seat of one's pants
4. Nonconscious execution: caught up in the moment, could not help oneself, could not control oneself, slipped out of one's mouth, do without thinking
5. Mental execution: using one's brain, put one's head to it, engage the thinking process, mental work
6. Reactive execution: reaction, reflex, deal with it as it happens, play it by ear, putting out fires
7. Repetitive execution: daily grind, bang away at, chip away at, spinning one's wheels
8. Iterative execution: cycle through, page through, cover each, touch on each
9. Periodic execution: do every so often, do from time to time, do regularly
10. Continuous execution: keep cranking out, keep on doing, do indefinitely
11. Triggered execution: do at the right moment, do when the time is right, do as orchestrated
12. Consecutive execution: do in order, follow the steps, do one thing at a time
13. Concurrent execution: do in parallel, juggle at the same time, multitask, do two things at once
14. Simultaneous execution: both do it at the same time, do at exactly the same moment
15. Collaborative execution: synergistic, in synchrony, pitch in, work as a team, combine forces, help each other out, lend a hand, act in unison, teamwork
16. Follow execution rules: walk a fine line, stay on the path, keep within bounds

4.25 EXECUTION CONTROL

Execution Control deals with the mechanisms of putting plans into action, and comprises two halves. First, Execution Control treats plans as sets of instructions and expectations about the effects the instructions will have on the world. It includes the concepts related to the launching and management of these instruction sets, including their starting, stopping, suspension, resumption, and continuation. Along with the basic control functions, this representation area includes concepts pertaining to the execution of particular types of plans with special specifications, including activities, performances, and skills – all of which may have plan representations that are not comprised of explicitly executable sets of instructions.

The best laid plans of mice and men often go awry. The second half of the job of Execution Control is to notice when and where this happens, and to proceed only as long as they are viable. Included here are judgments of when conditions and preconditions in plans are satisfied by the current world state, whether deadlines in plans are met or missed, and the consideration of variables relevant to plans as they are being executed, for example, the elapsed time, the portion of the plan yet to execute, and the accumulated costs of resources that have been expended. This area also includes

the key concept of an execution failure, the causes of which are the subject of a previous representational area, Causes of Failure.

1. Execute: launch, actuate, get to it, bring it on, get the show on the road, put it in motion, do it
2. Execute plan: carry out the plan, implement the plan, put the plan into effect
3. Execute explicit plan: carry out as instructed, do as specified, adhere to the plan, conform to the plan, following orders
4. Execute reusable plan: go through the motions, carry out as always, do the same as usual
5. Execute activity: participate in, take part in, engage in, been there done that
6. Execute performance: enact, reenact, perform, recite, exhibit, act out
7. Execute skill: work one's magic, display one's prowess, exhibit one's skills, put one's skills to work, show one's expertise
8. Suspend execution: pause, take a break, take a rest, hold up a moment, hold on a second
9. Suspended execution duration: lull, hiatus, sabbatical, vacation, time off, break time
10. Resume execution: jump back into, back to the job at hand, spin back up, pick up where one left off
11. Continue execution: carry on, continue forward, stay with it, hold steady, hang in there, stick with it, proceed with, never quit
12. Complete execution: wrap up, finish up, close the book, bring it home, put to bed, complete, conclude, close out, finalize, come to fruition, bring to an end, bring to a conclusion
13. Terminate execution: quit, bail out, jump ship, pull the plug, abort, throw in the towel, cease
14. Be distracted: lose focus, be sidetracked, preoccupied, pulled in a different direction, thrown off course
15. Execution rule: regulation, law, ordinance, restriction
16. Delayed execution: delayed start, running late, behind schedule
17. Currently executing plan: be up to, plan in progress, be in the middle of
18. Executed plan portion: progress, headway, work already done
19. Remaining plan portion: yet to be done, more to do, left on one's plate, loose ends, work left to do
20. Execution duration: as long as it takes, the time it took, period of performance
21. Execution start time: commencement, kick off, launch time, time of departure, showtime
22. Execution end time: wrap up time, time of completion, finish time
23. Miss deadline: late, tardy, past schedule, overdue, running behind, be behind on, miss the boat
24. Execution environment: stage, conditions, setting, arena, surroundings
25. Satisfied precondition: stage was set, groundwork, met requirements, ripe for the picking
26. Unsatisfied precondition: conditions not right, not be ready, jump the gun, do too soon
27. Violate precondition: become undone, undo progress, cancel out, unravel
28. Satisfied condition: met the requirement, sufficient, adequate amount, in line with

29. Unsatisfied condition: too little, insufficient, not up to par, inadequate amount, less than required
30. Constant across execution: invariant, steadfast, unaffected, unchanged by
31. Execution costs: running on empty, worth the effort, effortless, difficult, exhausting
32. Successful execution: pull it off, prevail, come to fruition, accomplish
33. Successfully executed plan: feat, achievement, deed, tour de force, stunt
34. Execution failure: botch, blunder, bungle, louse up, come up short, fail, crash and burn, awry
35. Consistently successful plan: standard approach, usual method, best practice, reliable scheme
36. Monitor execution: check progress, status check, watch what one is doing, see how things unfold, see it unfold

4.26 REPETITIVE EXECUTION

Repetitive Execution enumerates the subset of Execution Control concepts involved in the iterative and repetitive execution of plans. Here distinction is made between repetition, which involves doing the same thing over and over again until some condition is met, and iteration, which involves applying some action to every element in some set of definite size. In programming language terms, there is some analogy to the flow-of-control distinction between a while-loop and a for-loop. Here a repetition or iteration in the singular refers to the execution of a subplan that is embedded inside the loop, affording the plural concepts of repetitions or iterations that are already completed or yet to complete. Here we also include the mental events needed for counting repetitions and iterations, affording people both the ability to discover quantity of sets in the world via counting as well as the ability to execute actions a definite number of times.

1. Start repetition: begin repeating, start the cycle, commence orbit
2. Repeat: do again, loop, cycle again
3. Terminate repetition: break the cycle, abandon the routine, cease repeating
4. Increment count: chalk one up, mark one down, tally up
5. Repetitions count: tally, totals, number so far
6. Previous repetition: last time around, earlier cycle, prior loop
7. Current repetition: this time around, ongoing cycle, present loop
8. Next repetition: next time around, next on deck, next one up
9. Completed repetitions: finished cycles, past loops, ones in the bag
10. Remaining repetitions: future cycles, in the pipeline, rounds remaining
11. Start iteration: go through each one, start on the first, start going through
12. Iterate: take up the next one, do the next in line, continue through the series
13. Terminate iteration: quit before all are done, do as many as possible, leave some remaining
14. Complete iteration: exhaust the set, finish the series, no more to do

4.27 MIND–BODY INTERACTION

Mind–body interaction concerns commonsense psychology at its periphery, where mind and body meet. In the commonsense view the mind and the body are distinct,

yet have direct means of controlling and informing each other. The actions of the body, from moving one's hand to speaking some word, are seen as the result of some direction on the part of the mind. The directed action may fail or succeed, and in turn provides the mind with the feeling of its doing (body action perception), for example, as awkward or fluent, along with an array of other perceptions enabled by perceptual body parts, for example, one's eyes. These perceptions from the body share with the mind the resource that is one's focus of attention, with a scope that can be limited through concentration.

1. Mind: psyche, intellect, mental life, internal world, head, brain
2. Body: cadaver, carcass, corpse, physique, flesh and blood, figure, frame, remains, physical self, corporeal, earthly being
3. Body action: physical act, bodily movement, motion, gesture, move, budge
4. Body Action Direction: command one's body to move, try to move one's, try to position one's
5. Successful body action direction: gracefully, with dexterity, deftly, nimbly
6. Failed body action direction: awkwardly, one's body was not able, ungracefully, clumsy
7. Coordinated body action direction: align one's motions, move simultaneously, coordinate one's actions
8. Undirected body action: twitch, spasm, convulsion, unintentionally move, slip, stumble, trip, accidental movement
9. Reflex: involuntary response, automatic reaction, instinctively, out of instinct
10. Sense: perceptual organs, sensory modalities, sensory inputs
11. Perception: sensation, stimulus, feeling
12. Body action perception: feel oneself doing, the way it feels to do, sensation of doing
13. Awkward body action: to feel unnatural, strange sensation, uncomfortable action
14. Fluent body action: to feel natural, ergonomic, comfortable motion
15. Perceptual body action: tune out, close one's eyes/ears, shut out
16. Focus of attention: conscious of, on one's mind, in one's thoughts, in one's mind's eye
17. Attend: pay attention to, be aware of, be cognizant of
18. Concentrate: ignore everything but, attend only to, pay attention only to, focus only on
19. Mind-body state: state of mind, out of touch, in tune with, connected with one-self
20. Conscious state: alert, awake, wakeful, waking state
21. Altered state: feeling weird, to not feel normal, screwed up state, intoxicated, be high, inebriated
22. Unconscious state: unconscious, asleep, comatose, in a coma, in a trance, zombie-like

4.28 OBSERVATION OF PLAN EXECUTIONS

Observation of Execution relates to a special case of Plan Following, where a person is following a plan that is being executed by another person. In the general case, the ability to follow the actions of another person is a critical component of

collaborative execution, where two or more people attempt to execute a shared plan. However, many of the concepts in this representational area deal specifically with two more specific situations. First is the observation of another person's execution of some performance when the plan is known, for example, for the purpose of assessing their abilities. For example, an Olympic judge follows the performance of a gymnast as they perform their routine, and assesses how well they adhere to some idealized performance specification. Second is the observation of another person's execution when their plan is unknown, for example, for the purpose of learning how to do something. For example, novice surgeons will observe the performance of many surgeries during their medical residency in order to induce the complex plans that they will later execute themselves.

1. Observable execution: open behavior, visible action, outward expression
2. Unobservable execution: invisible action, hidden behavior, inner performance, behind the scenes, private thoughts
3. Observed execution: witness to, caught in the act, audience to, public display
4. Unobserved execution: behind closed doors, covertly, in secret, undetected, done in private, stealthily
5. Performance: recital, demonstration, exhibition, gig, presentation, ceremony, ritual
6. Performance specification: by the book, proper way, traditional way, accepted fashion, procedure, the ropes
7. Candidate performance specification: novice approach, beginner's intuition, assumed technique
8. Generate candidate performance specification: watch and learn, learn by example, mimic, get the hang of, get a feel for
9. Modify candidate performance specification: tweak one's method, adjust one's approach, change one's ways
10. Invalidate candidate performance specification: not see how one, fail to grasp how one, not get it
11. Validate candidate performance specification: catch on, have it down pat, to master, become proficient
12. Assessment: critique, check up, evaluation, judge, rank, assess
13. Assessment criteria: rubric, yardstick, standard, benchmark, baseline, litmus test
14. Assessment result: score, judgment, grade, after action review, progress report, criticism
15. Observe performance adherence: make sure it is done right, watch it being done, oversee, supervise
16. Observe performance divergence: catch a mistake, observe an error, notice a deviation

4.29 EMOTIONS

Emotions play a special role in the commonsense theory of the mind. Indeed, many might not view emotions as part of the mind at all. One might say, "Don't listen to your mind; listen to your heart," or "Don't think too much about it; trust your feelings." Likewise, it is common for people to ascribe emotional reasoning to a

spiritual component of their existence, distinct from their body and mind. This perceived dualism between emotional and nonemotional reasoning likely contributed to the widespread popular acceptance of the idea of emotional intelligence during the 1990s, with only tenuous links to psychological research (although see Salovey and Grewal, 2005). However, high-level explicit theories of the role of emotion in reasoning seem to play little or no role in characterizing the lexicon of emotion-related terms in the English language, which is largely dominated to references to specific emotional states.

The use of emotion terms has been studied extensively within the field of linguistics, with particular attention to cross-cultural differences. Good introductions to this line of cross-cultural linguistic research can be found in edited volumes by Athanasiadou and Tabakowska (1998) and Harkins and Wierzbicka (2001), and in the monograph of Wierzbicka (1999). Within computational linguistics there has been recent interest in creating large-scale text corpora where expressions of emotion and other private states are annotated (Wiebe, Wilson, and Cardie, 2005).

In our own analysis of language related to concepts of emotion, we were particularly struck by two characteristics of emotion vocabulary that distinguish them from those of other representational areas. First is the sheer quantity of single words that reference emotion states in the English language; there are hundreds of words that English-speakers use to describe how they are feeling. Second is the low level of polysemy within this set. There are lots of emotion words, and they are unambiguous. In the vocabulary list that follows, we also include feelings and other sensations, and cluster emotion concepts by how they relate to other representational areas of commonsense psychology.

Sensations, or things felt by your skin

1. Cold, Chilly, Frigid
2. Hot, Warm
3. Itchy, Tingling, Prickling, Tickling
4. Numb
5. Pain, Ache, Sore

Body feelings, or things felt mostly by your body

6. Hungry, Famished
7. Thirsty, Parched
8. Nausea, Sick, Disgust, Revulsion, Repulsed, Repugnance, Uncanny, Creepy
9. Aroused, Titillated, Lust, Lechery, Lasciviousness
10. Pleasure, Comfortable, Delight, Ecstasy, Rapture, Ravished
11. Displeasure, Uncomfortable, Discomfort, Awkward
12. Tearful, Weepy
13. Intoxicated, Inebriated, Drunk
14. Sedate
15. Dazed, Stupor
16. Calm, Serene, Peaceful, Tranquil, Relaxed, Repose
17. Stress, Tense, Overwhelmed
18. Tired, Exhausted, Fatigued, Weary, Sleepy
19. Energetic, Sprited, Exhilarated, Excited, Thrilled, Exuberant, Frisson

Knowledge state emotions

20. Certain, Confident, Sure, Conviction
21. Doubt, Uncertain
22. Confused, Bewildered, Perplexed, Puzzled, Disconcert, Discombobulate
23. Interest, Curiosity, Fascination, Wonder, Awe, Intrigue
24. Boredom
25. Amusement, Mirth, Jolly, Jocular, Entertained
26. Crazy, Deranged, Insane
27. Lost, Disoriented

Expectation emotions

28. Anticipation, Expectation, Suspense
29. Trust, Faith
30. Distrust, Suspicion
31. Startled
32. Surprised, Flabbergasted, Dumbfounded, Astonished, Amazed, Stunned
33. Shock, Consternation, Dismay, Stupefy, Appalled
34. Shaken, Rattled, Unnerved, Flustered, Ruffled, Fazed, Traumatized

Threat emotions

35. Fear, Frightened, Scared, Afraid
36. Hope, Encouraged, Optimistic, Sanguine
37. Anxiety, Nervous, Anxious, Worry, Concern, Unease
38. Dread
39. Panic, Frantic, Alarmed
40. Relief, Assuage, Alleviate
41. Terror, Horror
42. Uninhibited, Unrepressed, Unrestrained, Liberated, Free

Execution emotions

43. Apprehensive, Hesitant, Reluctance, Trepidation
44. Cautious, Careful
45. Courageous, Bold
46. Determined, Stubborn, Resolute
47. Eagerness, Enthusiastic, Gusto, Zeal, Zest, Passionate, Vigor, Vehement
48. Frustration, Exasperation, Aggravation
49. Intimidated
50. Shy, Diffident, Timid, Reserved, Self-conscious

Goal emotions, especially with respect to social relationships

51. Want, Desire, Crave, Longing, Yearn, Covet, Wish
52. Admiration, Worship, Reverence, Esteem
53. Respect
54. Humbled, Humility, Modesty
55. Devotion, Loyalty

- 56. Submissive, Meek
- 57. Envy, Jealousy, Covetous
- 58. Contempt, Disdain
- 59. Love, Infatuation, Adore, Agape, Amorous, Enamored
- 60. Affection, Fondness, Fancy, Friendliness, Endearment, Kindness, Tenderness
- 61. Aggressiveness, Belligerent, Bloodthirsty, Hostile, Bellicose, Confrontational, Combative
- 62. Hate, Loathe, Abhor, Animosity, Despise
- 63. Spite, Malevolence, Malicious, Malignant, Evil, Vicious

Goal achievement emotions

- 64. Happy, Pleased, Glad, Jubilant, Merry, Jovial, Cheerful
- 65. Joy, Elation, Euphoria, Glee
- 66. Grateful, Thankful, Appreciation
- 67. Gloating, Schadenfreude, Smug
- 68. Vindicated, Gratified
- 69. Pride, Proud, Conceit, Egotistic, Arrogance, Vain, Vanity

Goal failure emotions

- 70. Sadness, Unhappy, Displeased, Gloomy, Woe, Sorrow, Doleful, Glum, Wretched, Devastated
- 71. Disappointment, Heartbroken, Brokenhearted, Crushed
- 72. Disgruntled, Disquiet, Indignant
- 73. Suffer, Tormented, Misery, Agony, Anguish
- 74. Depressed, Despondent, Dejection
- 75. Despair, Hopelessness, Demoralized, Dispirited, Helplessness, Pessimism, Discouraged
- 76. Distress, Distraught, Upset, Bothered
- 77. Grief, Grief-stricken, Disconsolate, Mournful, Bereaved
- 78. Anger, Rage, Furious, Mad, Ire, Livid, Wrath, Outrage, Indignation
- 79. Annoyed, Irritated, Agitated, Aggravated, Perturbed, Vexed
- 80. Bitter, Resentment, Vengeful, Vindictive, Pique, Mad
- 81. Insulted, Offended, Umbrage, Hurt, Huffy
- 82. Consolation, Solace
- 83. Pity, Commiserate, Condole, Compassion, Mercy, Empathy, Sympathy
- 84. Lonely, Homesick
- 85. Alienated, Isolated, Estranged, Ostracized, Marginalized
- 86. Embarrassment, Chagrin, Humiliation, Discomfit, Foolish
- 87. Regret, Remorse, Repent, Rueful, Shame, Ashamed, Guilt

Goal management emotions

- 88. Satisfied, Satiated, Content
- 89. Discontent, Restless, Restive
- 90. Acceptance, Resignation
- 91. Carefree, Lighthearted, Gay
- 92. Playful, Whimsy

Decision emotions

- 93. Like, Inclination, Partiality, Penchant, Predilection, Preference, Bias towards, Taste for
- 94. Dislike, Disinclination, Aversion, Distaste, Detest
- 95. Conflicted, Torn
- 96. Indifferent, Disinterested, Ambivalent, Unconcerned, Apathy

Monitoring states feelings

- 97. Focused, Attentive, Alert
- 98. Mesmerized, Transfixed
- 99. Distracted

Observation of execution emotions

- 100. Approval
- 101. Disapproval, Reproach, Deprecate
- 102. Impressed

Emotion emotions

- 103. Emotionless, Unmoved, Untroubled, Aloof

Although we do not consider each of these words in Chapter 49 on formalizing emotions, we deal with a large enough illustrative subset of them to give a good idea about how the others would be handled.